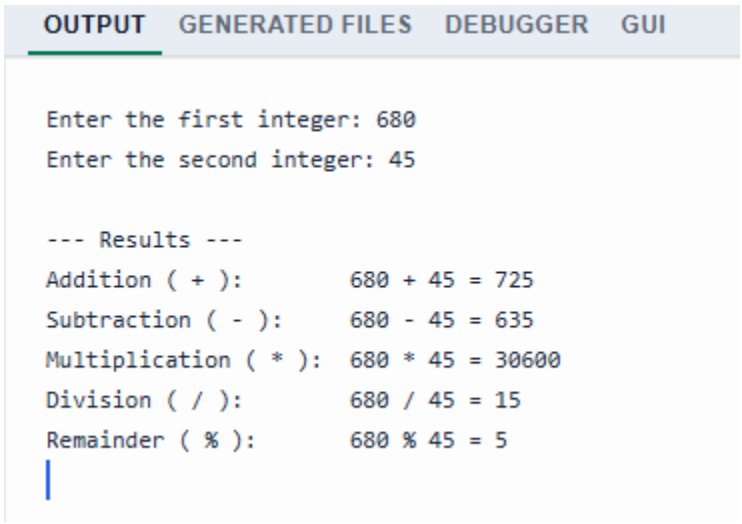
```
import java.util.Scanner;

public class FourFunctionCalculator {
    public static void main(String[] args) {
        // Create a Scanner object to read user input
        Scanner scanner = new Scanner(System.in);

        // 1. Read two integers from the user
        System.out.print("Enter the first integer: ");
        int num1 = scanner.nextInt();

        System.out.print("Enter the second integer: ");
        int num2 = scanner.nextInt();

        System.out.println("\n--- Results ---");

        // 2. Print the result of basic arithmetic operators with labeled output
        System.out.println("Addition ( + ):        " + num1 + " + " + num2 + " = " + (num1 + num2));
        System.out.println("Subtraction ( - ):     " + num1 + " - " + num2 + " = " + (num1 - num2));
        System.out.println("Multiplication ( * ):  " + num1 + " * " + num2 + " = " + (num1 * num2));

        // Check to prevent division by zero runtime errors
        if (num2 != 0) {
            System.out.println("Division ( / ):        " + num1 + " / " + num2 + " = " + (num1 / num2));
            System.out.println("Remainder ( % ):       " + num1 + " % " + num2 + " = " + (num1 %
num2));
        } else {
            System.out.println("Division ( / ):        Cannot divide by zero.");
            System.out.println("Remainder ( % ):       Cannot divide by zero.");
        }

        scanner.close();

        /*
         * 3. REQUIREMENT: NOTE ON NEGATIVE OPERANDS
         * In Java, the Division (/) and Remainder (%) operations behave uniquely with negative
numbers:
         *
         * - Division (/): Truncates toward zero instead of rounding down to the lower integer.
         *   For example, -5 / 2 equals -2 (not -3).
         *
         * - Remainder (%): The sign of the result matches the sign of the FIRST operand (the
dividend).
         *   The sign of the second operand (the divisor) is ignored completely.
```

```
        *   For example, -5 % 2 equals -1, while 5 % -2 equals 1.
        */
    }
}
```